

02285 AI and MAS, F26

Theory Assignment 2026: Paper Presentation

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1 Goals of the assignment

In this assignment, you are asked to find a research paper relevant to the course curriculum (and potentially also the programming project), and make a video presentation of it. The goals of the assignment are:

- Practice searching for relevant research literature.
- Practice reading and understanding research papers.
- Practice presenting the core contents of research papers in a clear and concise way, and critically analyse and review them.

2 General requirements

2.1 Group sizes and group work

This is as for the warmup assignment. Groups of 2 can make a video of up to 6 minutes, and larger groups will have +2 minutes per extra group member, so:

- **Group size 2:** max 6 minutes video.
- **Group size 3:** max 8 minutes video.
- **Group size 4:** max 10 minutes video.
- **Group size 5:** max 12 minutes video.

We of course expect that larger groups can do more thorough research for a relevant paper, and give a more thorough presentation and analysis of the paper.

2.2 Collaboration policy

We encourage you *not* to discuss your choice of paper or the content of your presentation with other groups in order not to influence each other unnecessarily. However, you are still allowed to do it, as long as you in the beginning of the video declare which other groups you talked to and exactly about what.

2.3 Use of generative AI

It's not prohibited to use generative AI tools such as LLMs to assist in finding relevant papers, and to assist in summarising them. However, the final presentation should be your own, based on your own reading and understanding of the paper. If you ask an LLM to describe an algorithm, it will often simplify it and leave out crucial technical details, which can be problematic. Any use of generative AI tools should be clearly declared in the beginning of the video.

2.4 Handing in

Your solution is to be handed in via DTU Learn. The hand-in is a single movie file in a standard video format (e.g. mp4 or mov). Requirements:

- **Length.** It has to fit the length requirements stated in Section 2.1.
- **Declarations.** You need to start the video by a group declaration following the same instructions as for the warmup assignment. After this, you should declare any collaboration with other groups (see Section 2.2), and any use of generative AI tools (Section 2.3).
- **Content and style.** You should still follow the general requirements in the document `guide_for_video.pdf`, though of course here you will be presenting the theories and algorithms of someone else, not your own. Crucially, *don't read from a script*.

3 Details of the assignment

As mentioned, the assignment is to find a relevant research paper, and make a video presentation of it.

3.1 Conditions for relevance

A relevant paper is any research paper that relates to the course curriculum in a non-trivial way. It can for instance be a paper that extends or applies one (or more) of the theories or algorithms covered in the course. You are very welcome to search for papers that are relevant to the programming project, so that your work on the theory assignment will also give you a head start on the programming project. However, this is not a requirement. It is however a requirement that the paper connects to the course curriculum in some obvious way.

- **Year of publication.** More recent papers are often more interesting and relevant, but this is definitely not a rule. The team winning the programming competition in 2025 based their solution on a master thesis at DTU that the student presented in a guest lecture in the course. The master thesis defines a new version of conflict-based search (CBS) using jump point search (JPS). The jump point search algorithm is from 2011, but then turned out to still be highly relevant as it could significantly speed up CBS. The master student also wrote a research paper about this (currently under review).
- **Venue of publication.** The paper should be peer reviewed and published in an international AI journal or conference. Almost all AI conferences and journals have peer review, so if it's published in any of those, it should be OK. You should only under special circumstances

allow yourself to use a paper that is not yet peer reviewed and published, such as a very recent preprint on Arxiv, as it is much harder to guarantee the quality of such papers (note that published papers might also be on Arxiv, it's only a problem if they are *only* on Arxiv).

3.2 Finding a relevant paper

There are many ways to search for relevant papers. Sometimes a regular search engine can be helpful when just providing it with relevant keywords. However, often specialised search engines such as Google Scholar are more helpful.

In Google Scholar, one can also search for papers that cite a given paper. This can be very helpful, e.g. if you'd like to see which recent papers have built on top of some of the theories and algorithms covered in the course, like CBS or IW. Then you can just find the original paper describing the theory or algorithm, and then search for papers that cite it.

When you search, you can either take departure in a particular problem domain, like multi-agent pathfinding (MAPF), a particular formalism, like PDDL, or a particular algorithm, like IW. It's recommended that you first try to think about what you found most interesting in the course curriculum, and base your search on that. The more interesting and relevant you find the paper you end up choosing, the more interesting your presentation will be, and the more you will learn from it.

You are also very welcome to orient your search towards papers that present ideas that might be helpful for you in the programming project, as long as they also relate specifically to the course curriculum in one way or another. If you want to find such papers, you might first want to brainstorm a bit about which approaches you would like to pursue in the programming project (the assignment text of the programming project is available on DTU Learn).

3.3 What to include in the presentation

In your presentation, in addition to the declarations mentioned in Section 2.4, you should cover the following (not necessarily in that order):

- The paper's title, authors, publication venue and year.
- The main problem the paper addresses.
- The key contributions and innovations of the paper.
- How the paper relates to concepts, theories and algorithms covered in the course.
- If relevant, how you see the paper could be helpful to finding strong algorithms for the hospital domain (mainly multi-agent levels that you are going to work exclusively with in the programming project).
- What made you choose this exact paper?
- A precise walk-through of the theory and/or algorithms presented in the paper, including any relevant examples. If the paper presents novel definitions or algorithms, these should be included and explained in your own words in the presentation. Your video is supposed to address an expert audience, so you should not shy away from including technical details. Of course bigger groups will have time and resources to dig a bit deeper into the technical details.
- A critical analysis of the paper: How valuable do you find the contribution, do you agree with the authors' claims, can you identify any important limitations or weaknesses of the paper?

If you don't have time to cover all details of the algorithms or theories of the presented paper, it's very important that you make it clear which parts or details you have left out.